

PHILADELPHIA INCLUSIONARY ZONING

Mixed Income Neighborhoods Impact Analysis

Evaluating the effect of Philadelphia's MIN overlay on residential development

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OVERVIEW

What This Presentation Covers

This presentation moves from the policy to the evidence to what it means for Philadelphia:

- **Policy context** — Philadelphia's affordability pressures, how the MIN overlay works, and the debate over whether it acts as an implicit tax on development.
- **Research design** — three research questions, the L&I permit data, and a two-stage descriptive-then-causal analytical strategy.
- **Findings** — evidence on permitting volume, project size, and the mix of housing types built.
- **Synthesis & recommendations** — what the descriptive and causal results reveal together, and the takeaways for policy.

GUIDING QUESTION

Did the MIN overlay suppress residential development in its first four years?

A two-stage approach:

1 Descriptive analysis of permitting trends inside vs. outside the overlay.

2 A difference-in-differences test matching MIN tracts to comparable non-MIN tracts.

BACKGROUND

Addressing Affordability Concerns

51.7%

of Philadelphia households are rent-burdened, spending >30% of income on housing

20.3%

poverty rate — among the highest of major cities in the Northeast

Dual Goals

Promote growth while preserving affordability for existing residents

THE ROLE OF ZONING

- Zoning is the primary tool cities use to shape land use.
- In Philadelphia, projects meeting the code advance “by-right,” while those that do not must seek a variance before the Zoning Board of Adjustment.
- Variance hearings can add months significant costs, so developers often adapt projects to avoid a ZBA hearing.

TWO DESIGNS FOR INCLUSIONARY ZONING

- Opt-in bonuses — projects receive relaxed requirements (e.g., added height) in exchange for reserving affordable units.
- Mandatory overlays — hard affordability requirements imposed on strategic neighborhoods. The MIN overlay is this type.

Source: Pew Charitable Trusts, 2025 State of the City (Philadelphia).

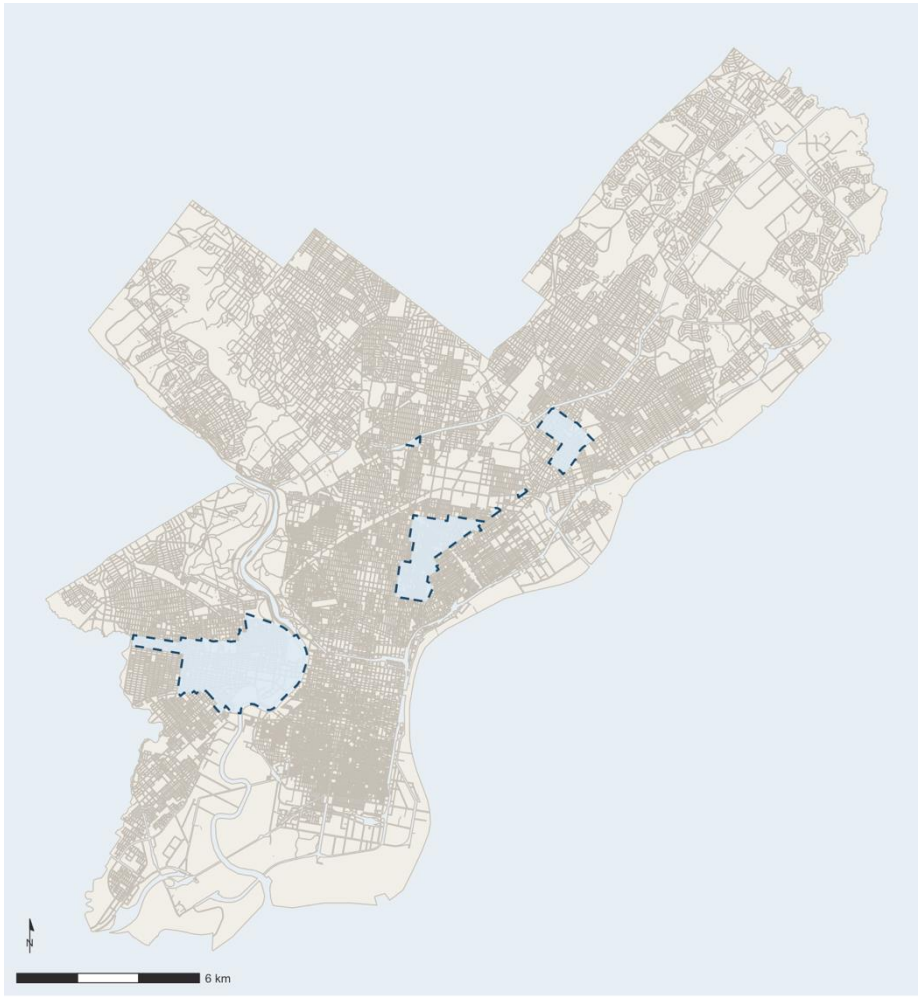
POLICY CONTEXT

The Mixed Income Neighborhoods Overlay

- Enacted by Philadelphia City Council across six areas of the city.
- Applies to new construction or substantial rehabilitation with 10 or more residential units.
- Requires 20% of units to be reserved at rents affordable to households earning up to 40% of area median income (AMI).
- Designed to preserve affordability on the periphery of Philadelphia's fastest-growing neighborhoods.

Unit type	Rent @ 40% AMI	HUD Fair Market Rent
Studio	$803	$1,303
1 Bedroom	$860	$1,451
2 Bedroom	$1,032	$1,737
3 Bedroom	$1,193	$2,100

Affordable vs. market rents in Philadelphia, 2024.



MIN overlay district boundaries, Philadelphia

The Debate: An Implicit Tax on Development?

- Economic theory treats mandatory inclusionary zoning as an implicit tax on new development.
- The policy has few supporters among developers; one active within the MIN zones called it the “death knell” for local residential development.

SUPPLY-SIDE THEORY PREDICTS THREE DEVELOPER RESPONSES

1. **Reduce total starts**
   Build fewer projects overall in response to the added cost.

2. **Bunch below the threshold**
   Shift toward projects under 10 units to avoid the requirement.

3. **Enlarge qualifying projects**
   Scale up to spread compliance cost across more market-rate units.

ANALYTICAL STRATEGY

Three Research Questions

Q1

Did the overlay depress permitting volume?

Total volume of new residential construction inside the overlay.

Q2

Did developers shrink projects or group below the 10-unit threshold?

Project size and clustering just under the trigger.

Q3

Did the composition of new projects change?

The mix of dwelling types: single-family, townhouse & multi-family.

Two-stage approach: each question is examined descriptively, then tested for causality with a difference-in-differences design.

APPROACH

Data & Study Design

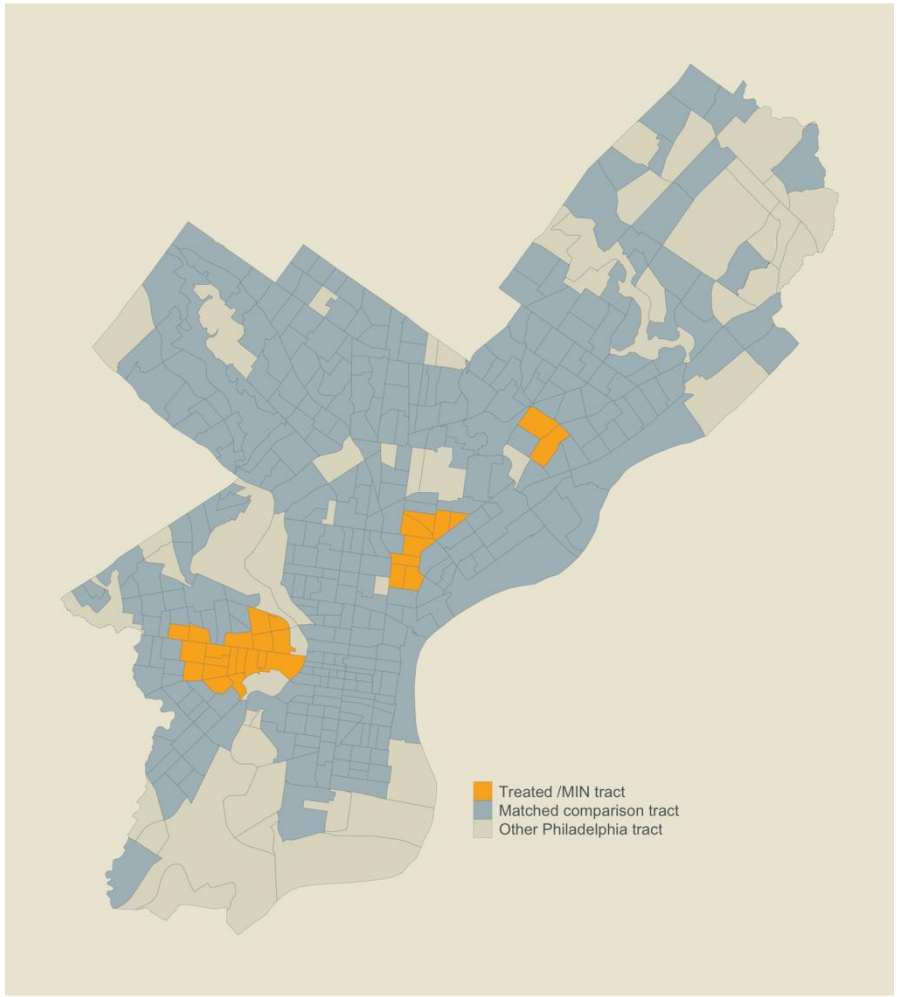
- Data: Philadelphia L&I permit records (2015–2026) linked to Census tracts.

STAGE 1 — DESCRIPTIVE TRENDS

- Compare permit volume, project size, and dwelling-type composition inside the overlay against the rest of the city.

STAGE 2 — DIFFERENCE-IN-DIFFERENCES

- 29 treated MIN tracts (≥50% of land in the overlay) matched to comparable non-MIN tracts via propensity-scores:
    - Measures similarities along income, tenure, vacancy, CMX land share, and pre-2022 permitting.
- Two-way fixed-effects event study and pooled DiD with standard error corrections.



29 treated MIN tracts matched to weighted comparison tracts across Philadelphia.

FINDING 1 OF 3

Q1 — Did the MIN Depress Permitting Volume?

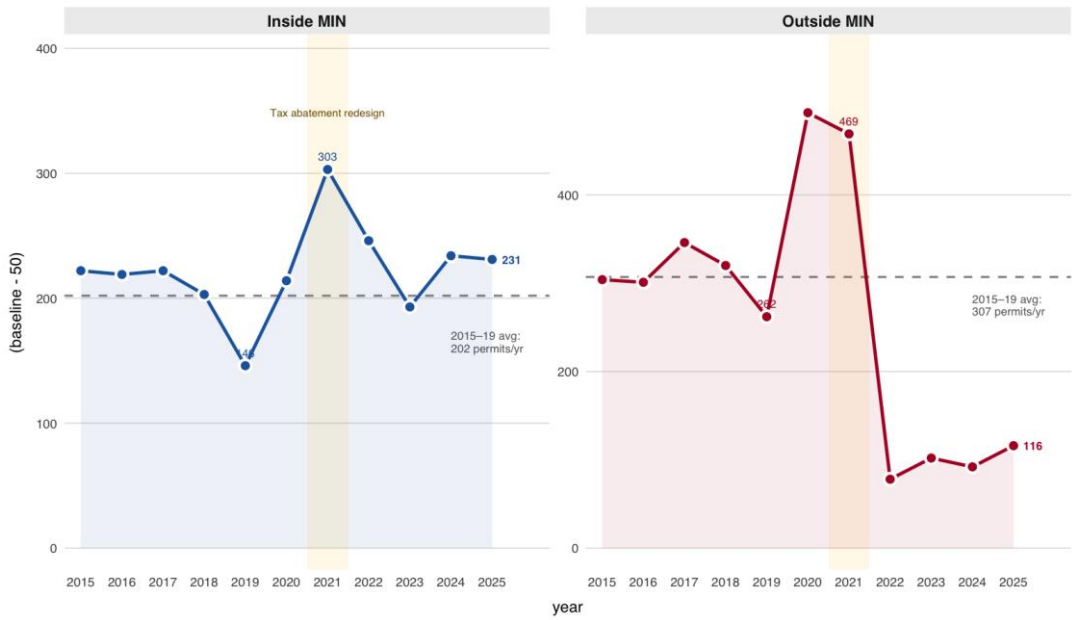
DESCRIPTIVE EVIDENCE

- The inside/outside CMX permit ratio reversed from ~0.66 (2015–19) to 1.99 (2025).
- From 2021→2022, CMX new construction fell 18.8% inside the overlay but 83.4% outside.

CAUSAL TEST

- Within the matched neighborhoods, there is no effect– All results are statistically insignificant.
- Insignificance is stable across 3 robustness checks:
    - threshold, COVID donut, effective-date shifts

VERDICT: NO The evidence doesn't suggest the overlay depressed permitting volume.



CMX new-construction permits: inside vs. outside the MIN overlay, 2021–2025. Both series fell from a 2021 abatement-driven peak.

FINDING 2 OF 3

Q2 — Did Developers Shrink or Bunch Below 10 Units?

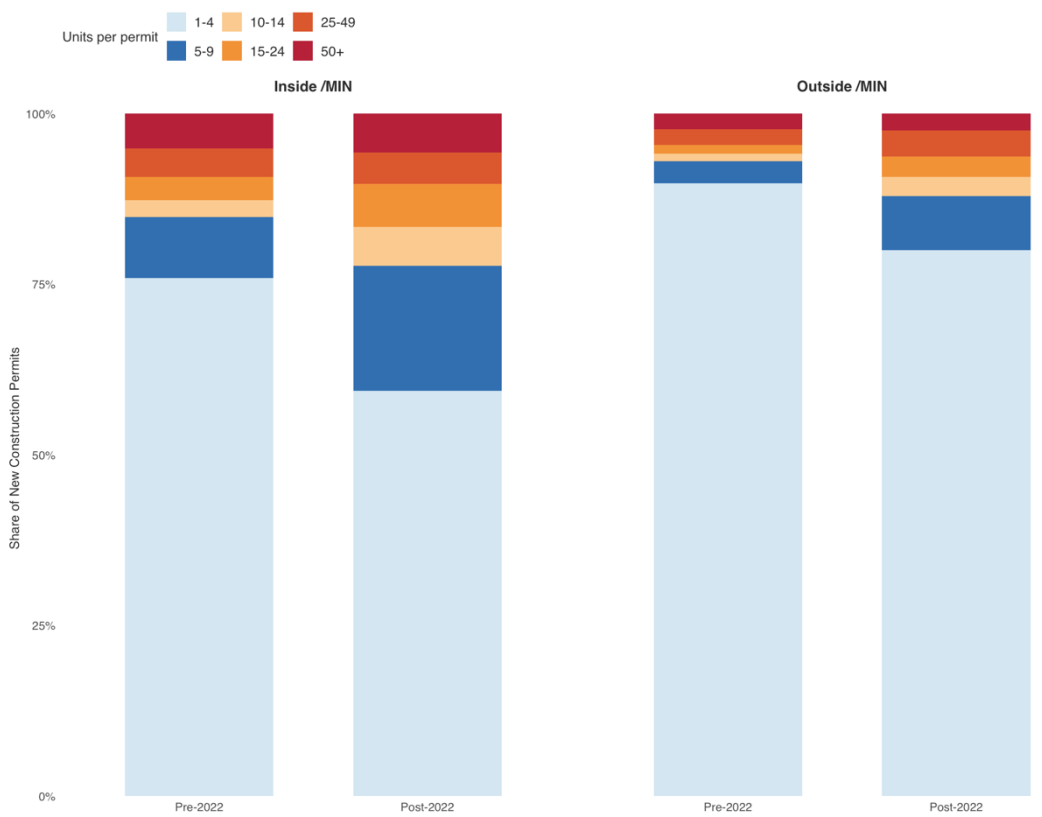
DESCRIPTIVE EVIDENCE

- Average MIN project size rose from 6.21 to 16.74 units; the 5–9-unit share (just below the trigger) roughly doubled, 8.9% → 18.3%.
- But the 5–9-unit band also grew citywide (3.2% → 7.9%) — not unique to the overlay.

CAUSAL TEST

- Dedicated 5–9-unit bunching outcomes are insignificant in both families.
- The only significant estimate points the opposite way (+27% units/permit, all-parcel) and is fragile (CR2 p = 0.053).

VERDICT: NO Evidence of threshold avoidance or shrinkage isn't robust.



Unit-count share of new construction permits, inside vs. outside the MIN, pre- vs. post-2022. The 5–9-unit bin (dark blue) grew in both.

FINDING 3 OF 3

Q3 — Did the Composition of Projects Change?

DESCRIPTIVE EVIDENCE

- Inside the MIN, single-family fell 42.8% → 5.7%, townhouse rose 31.8% → 69.3%, and multi-family rose 15.9% → 23.5%.
- But the single-family decline and townhouse rise mirror citywide trends — only multi-family shifts more inside the overlay.

CAUSAL PERSPECTIVE

- Composition is the one question the DiD design cannot test directly given the tract-parcel data structure.
- The multi-family excess shift remains descriptive
- The most plausible true Mixed Income Neighborhoods effect and the best lead for future causal work.

VERDICT: MAYBE multi-family growth is the clearest MIN-specific signal, not able to be confirmed causally.

Composition change, MIN minus non-MIN (percentage points)

Dwelling type	MIN Δ	Non-MIN Δ	Excess	MIN-specific?
Single-family	−37.1	−39.9	+2.8	No
Townhouse / row	+37.5	+43.4	−5.9	No
Multi-family	**+7.6**	**+0.3**	**+7.3**	**Yes**
Two-family / duplex	−4.8	−2.3	−2.5	No

Only multi-family construction grew meaningfully more inside the overlay than in the rest of the city.

SYNTHESIS

What the Two Analyses Reveal

Descriptive signals that look striking in isolation shrink toward zero once the comparison group is properly matched.

- The descriptive stage surfaced three apparent MIN effects: a post-2022 volume slowdown and CMX ratio reversal, a doubling of the 5–9-unit share, and a shift in dwelling-type composition.
- The causal stage tested the first two and found them to be largely citywide artifacts. The raw inside/outside reversal reflected comparison to dissimilar tracts more exposed to the 2021 abatement boom and the post-2022 financing squeeze — not the overlay itself.
- MIN tracts are lower-income, with higher renter shares, vacancy, and a different pre-period permitting trajectory. Suppression effects appear only when the comparison includes dissimilar Philadelphia; once matched, they disappear.
- The clearest MIN-specific footprint is on the composition of new projects — disproportionate multi-family growth — not on their volume or threshold positioning. It is best treated as a lead for future research as more post-2022 data accumulates.

CONCLUSION

Policy Takeaways & Recommendations

KEY TAKEAWAYS

- In its first four years, the overlay shows no distinct effect on the volume, size, or threshold-positioning of new residential construction.
- Developer concerns of suppressed starts or bunching below the 10-unit trigger are not supported once compared to a credible counterfactual.
- The post-2022 slowdown is better explained by the 2021 tax-abatement redesign and the inflationary cycle than by the overlay specifically.
- The requirement may simply be small enough not to deter development — or it is too early to determine, given the lag between permitting and move in.

RECOMMENDATIONS

1

Maintain the overlay as-is

The evidence does not support repeal or rollback on development-suppression grounds; the feared chilling effect does not appear in the matched estimates.

2

Re-evaluate on a longer horizon

Re-run the difference-in-differences analysis as more post-2022 permit data accumulates, with particular attention to multi-family composition.

Difference-in-Differences: Pooled Estimates

Outcome (family)	Estimate	Plain reading	p (clustered / Conley)
Units per permit (CMX)	+0.15 log	~+16%, n.s.	0.69 / 0.57
Permit count (CMX)	+0.03	~+3%, n.s.	0.95 / 0.96
5–9-unit permits (CMX)	+0.58	n.s.	0.44 / 0.57
Units per permit (all)	**+0.24 log**	~+27%	**0.037 / <0.001**
Permit count (all)	−0.21	~−19%, n.s.	0.54 / 0.50
5–9-unit permits (all)	+0.16	n.s.	0.70 / 0.74

Reading the table. Every estimate is statistically insignificant except units-per-permit (all parcels), which is fragile under the small-sample correction (CR2 p = 0.053) and points toward larger — not smaller — projects. Descriptive signals shrink toward zero once the comparison group is matched.

Two-way clustered SEs (tract + year-month); n.s. = not significant. Outcome family: all new construction.